Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
  Black: Molex 2147601225
  Red: Molex 2147602225

Reverse Polarity Protection

Q13
CSD19534Q5A

J10
POWER
V_IN
V_IN_RTN
Molex
0436500200

D1
SMDJ24CA
24V

C57
10uF-50V-10%-X7R-1210

C55
0.1uF-50V-10%-0603

U7
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C48
10uF-50V-10%-X7R-1210

R28
5.1k-1%-0603

R30
5.1k-1%-0603

R25
5.1k-1%-0603

D2
PWR

VIN_PS

File: buck-boost-power-supply.kicad_sch
2/5
+V_IN
-V_IN
15V 3A Buck-Boost Power Supply
+V_OUT
-V_OUT

Power Supply Back
Feed Protection

Q1
CSD19534Q5A

C1
10uF-50V-10%-X7R-1210

C13
0.1uF-50V-10%-0603

U1
LM74700QDBVRQ1
VCAP
EN
GND
A
G
C

C14
10uF-50V-10%-X7R-1210

Mating Connector:
Housing: Molex 0436450200
Precrimped 18AWG wire 450mm:
  Black: Molex 2147601225
  Red: Molex 2147602225

J4
V_OUT
V_OUT_RTN
15V
Molex
0436500200

Mating Connector:
Housing: Molex 5040510401
Precrimped 24AWG wire 150mm: Molex 0797581003

J7
USB1
Molex
5040500491

File: canable2.kicad_sch
3/5
CANable2.0
USB_DP
USB_DM
3V3_USB
GND_CAN
CAN_HI
CAN_LO

J9
CAN1
MFG P/N: 0705550036
MFG: Molex

J1
Terminal

J2
Terminal
Keystone Electronics
7774

File: NX-Button-Interface.kicad_sch
4/5
NX-Button Interface

J3
NX CONTROL & UART
3V3_BTN
NX_PC_LED-
NX_PC_LED+
UART2_RXD
UART2_TXD
AC OK
Auto Power On Disable
SYS_RESET
REC
GND
SLEEP/WAKE*
Molex
PH1RB-15-UA
GND1

3V3_BTN
UART2_TXD
UART2_RXD
SYS_RESET
REC
NX_PC_LED+
NX_PC_LED-

D3
PC

SW2
RECOVERY

SW1
RESET

GND1

File: usb-serial.kicad_sch
5/5
USB Serial
3V3
RXD
TXD
RESET
FORCE_RECOVERY
UART2_TXD
UART2_RXD

H3
MountingHole
H10
MountingHole
H11
MountingHole
H4
MountingHole

H5
MountingHole
H7
MountingHole
H8
MountingHole

H1
1.152mm ToolingHole
H12
1.152mm ToolingHole
H2
1.152mm ToolingHole
H13
1.152mm ToolingHole

Light Pipe
LP1
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP2
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP3
Bivar
Bivar-SLP3-150-100-F

Light Pipe
LP4
Bivar
Bivar-SLP3-150-100-F

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Sheet: /
File: TVS-Adapter-v1.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-17	Rev: 1
KiCad E.D.A. 10.0.7		Id: 1/5

+V_IN
−V_IN
+V_OUT
−V_OUT
GND
Vcc

C8 47uF 50V
C12 47uF 50V
C47 10uF−50V−10%−X7R−1210
C10 10uF−50V−10%−X7R−1210
C9 10uF−50V−10%−X7R−1210
C7 10uF−50V−10%−X7R−1210
C11 10uF−50V−10%−X7R−1210
C46 10uF−50V−10%−X7R−1210
C52 10uF−50V−10%−X7R−1210
C53 10uF−50V−10%−X7R−1210
C6 10uF−50V−10%−X7R−1210

HDRV1 Q12 CSD19534Q5A
LDRV1 Q11 CSD19534Q5A
Q5 CSD17578Q5A HDRV2
Q8 CSD17578Q5A LDRV2
SW1
SW2
L1 2.2uH−19A−5.1mohm−20%

RCS
CSG
R20 10mOhm−3W−1%−2512

C17 10uF−50V−10%−X7R−1210
C21 10uF−50V−10%−X7R−1210
C20 10uF−50V−10%−X7R−1210
C16 10uF−50V−10%−X7R−1210
C26 10uF−50V−10%−X7R−1210
C25 10uF−50V−10%−X7R−1210
C2 10uF−50V−10%−X7R−1210
C4 10uF−50V−10%−X7R−1210
C5 10uF−50V−10%−X7R−1210
C3 10uF−50V−10%−X7R−1210

D9 BAT54KFILM
R29 249K−1%−0603
R26 93.1k−1%−0603
R23 2K−1%−0603
R27 10R0−1%−0603
C41 0.1uF−50V−10%−0603

U4 Texas Instruments LM5176PWPR
VIN
VISNS
EN/UVLO
SS
COMP
MODE
RT
DITH
SLOPE
CS
CSG
VCC
BIAS
VOSNS
HDRV1
BOOT1
SW1
LDRV1
FB
PGOOD
HDRV2
BOOT2
SW2
LDRV2
ISNS(+)
ISNS(−)
AGND
PGND
EP

C40 0.1uF−50V−10%−0603
R5 357K−1%−0603
R9 20K−1%−0603
D8 BAT54KFILM
C39 1uF−50V−10%−0603
C42 0.1uF−50V−10%−0603
D6 BAT54KFILM
C31 0.1uF−50V−10%−0603
R31 0R−1%−0603 HDRV1
R24 0R−1%−0603 LDRV1
R16 0R−1%−0603 HDRV2
R19 0R−1%−0603 LDRV2

C35 15nF−10%−50V−0603
C28 5.6nF−10%−50V−0603
R10 5.9k−1%−0603
C30 5.6nF−10%−50V−0603
C33 270pF−10%−50V−0603
R22 205K−1%−0603
R21 13K−1%−0603
C37 150pF−50V−10%−0603
C29 47pF−50V−10%−0603
R12 100R−1%−0603
R11 100R−1%−0603

VIN(MAX)	24V
VIN(MIN)	5V
VOUT	15V
IOUT(MAX)	3A
FSW	589 kHz
Efficiency	96.7%

15V−3A−Power Supply Analysis

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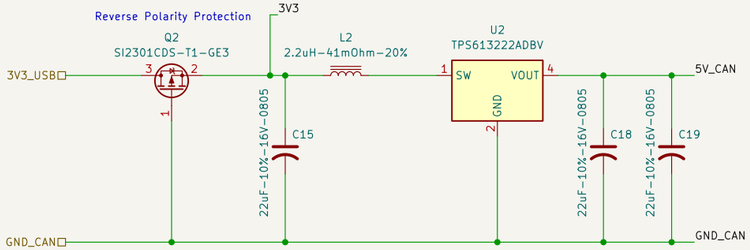
Sheet: /15V 3A Buck−Boost Power Supply/
File: buck−boost−power−supply.kicad_sch

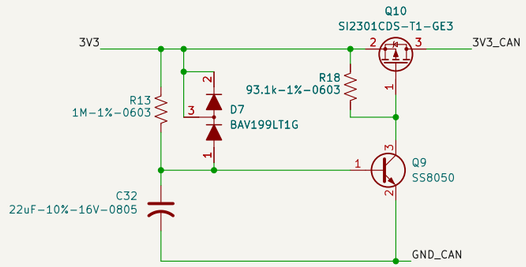
Title: The Vision System Adapter

Size: B	Date: 2026−10−17	Rev: 1
KiCad E.D.A. 10.0.7		Id: 2/5

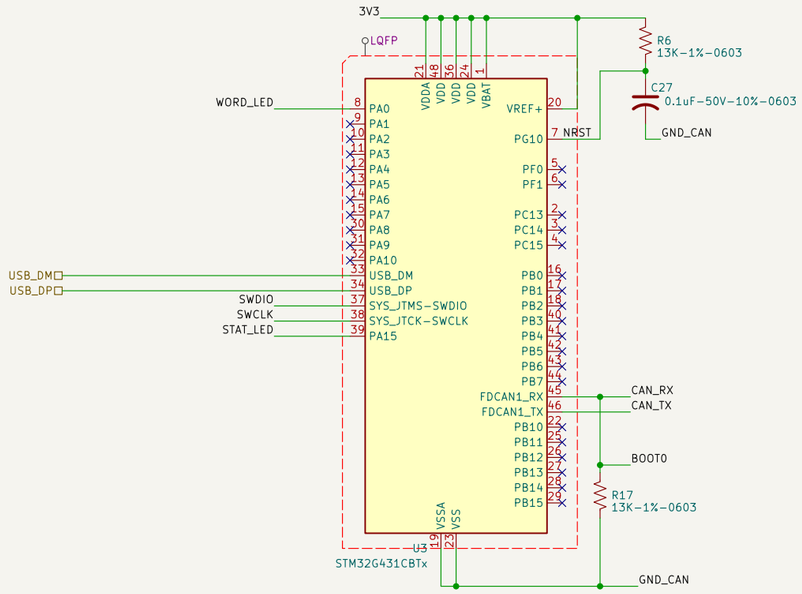
5V-250mA-Power Supply Analysis

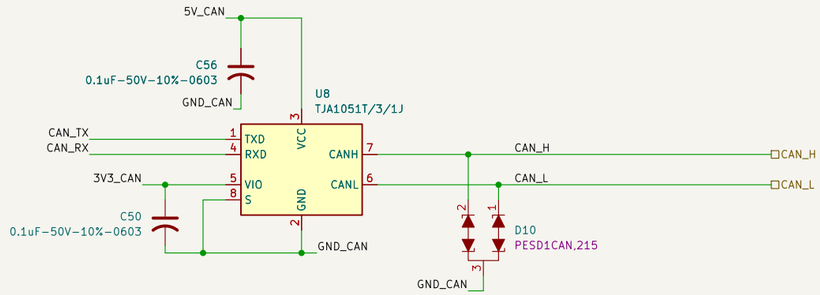
Reverse Polarity Protection

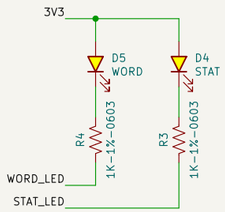




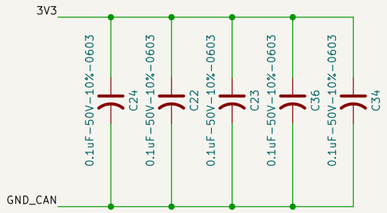
3V3_CAN delays bringing CAN RX (BOOT0) high until NPN turns on. ~10sec.











CANable 2.0 Schematic

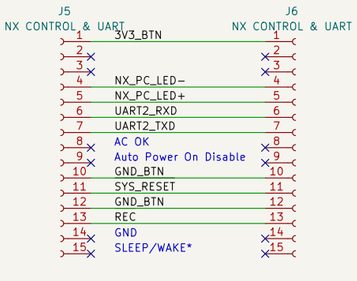
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Sheet: /CANable2.0/
File: canable2.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-17	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 3/5



H6
M3.7 MountingHole

H9
M3.7 MountingHole

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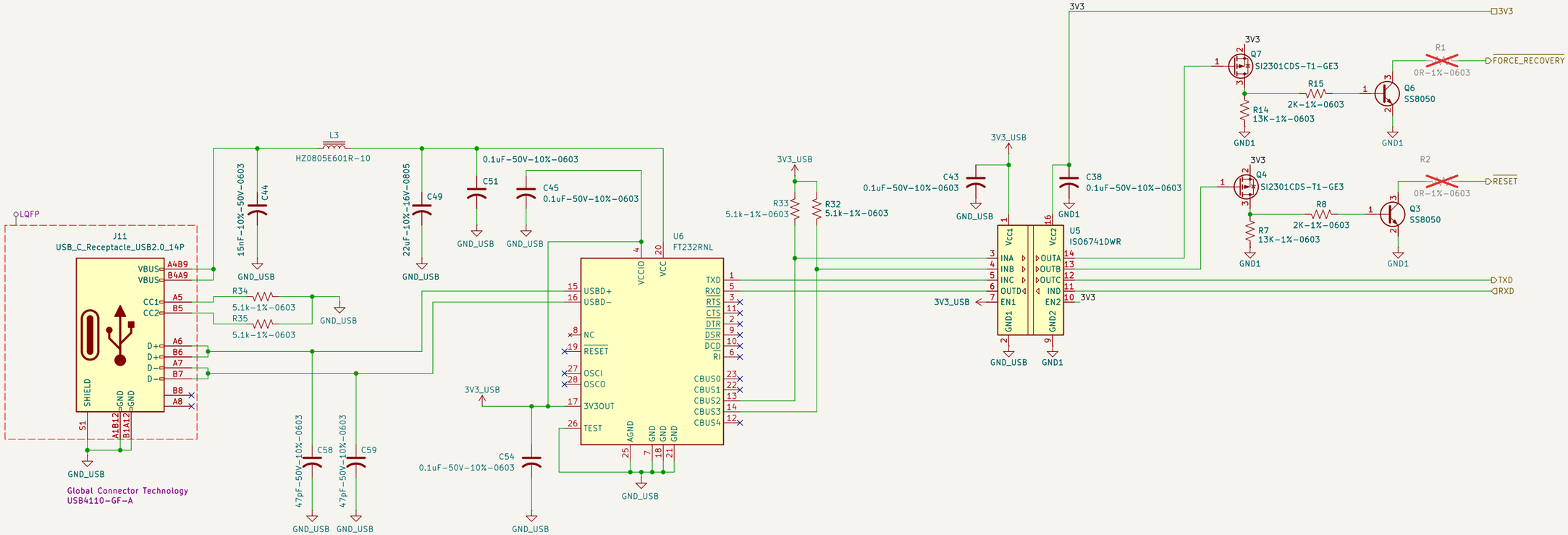
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Sheet: /NX-Button Interface/
File: NX-Button-Interface.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-17	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 4/5



https://hackaday.com/2023/08/07/all-about-usb-c-example-circuits/

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Sheet: /USB Serial/
File: usb-serial.kicad_sch

Title: The Vision System Adapter

Size: B	Date: 2026-10-17	**Rev: 1**
KiCad E.D.A. 10.0.7		Id: 5/5